Phenotyping with AI Agents

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Abstract. Understanding how intelligent agents make decisions under uncertainty is central to Artificial Intelligence (AI) evaluation. This study profiles Gemini and DeepSeek in the Iowa Gambling Task and compares them with human participants across matched task variants. We combine model-free behavioural features with Rescorla–Wagner (RW) modelling, Principal Component Analysis (PCA), clustering, and test whether prompt-engineered personas induce distinct phenotypes. Both LLMs exhibit structured learning rather than random behaviour but stand apart from humans in important ways. Gemini achieves the highest performance and is most strongly captured by the RW model. DeepSeek shows lower and more variable performance but indicates a phenotype distribution closer to humans. Persona prompting shifted agents between existing behavioural patterns rather than creating entirely new ones, with only the risk-seeking persona forming a clearly distinct profile. These findings suggest that behavioural phenotyping can reveal decision-making patterns hidden by performance metrics alone and may support safer evaluation of AI systems in uncertain environments.

Keywords: Large Language Models · Behavioural Phenotyping · Iowa Gambling Task · Rescorla–Wagner Modelling · Sequential Decision-Making · Human-AI Mapping

1 Introduction

1.1 Motivation and Background

AI systems increasingly make sequential decisions under uncertainty. Despite dramatic advances in model performance, the field still lacks a clear understanding of how and why these systems behave the way they do [1]. Two systems can achieve similar scores through very different strategies [2].

Unlike traditional reinforcement learning (RL) algorithms, Large Language Models (LLMs) are not explicitly trained to optimise reward in sequential environments, yet they are increasingly being used in settings that require decision-making [3, 4]. For safe deployment, it is important to understand whether their decision-making strategies align with human psychology [5]. Therefore, this paper evaluates LLM behaviour through computational cognitive modelling rather than surface-level output quality.

1.2 Why is the IGT a Good Testbed?

The Iowa Gambling Task (IGT) is an ideal testbed designed to investigate decision-making behaviour by simulating how agents form profitable long-term strategies under conditions of uncertainty and competing reward structures [6]. As the rules and payout probabilities are unknown to a player, the task forces them to learn continuously from sequential feedback. The IGT also captures the trade-off between exploration and exploitation and isolates sensitivity to rewards and penalties. Thus, the IGT is well suited for studying risk preferences and adaptive learning in both humans and LLMs.

1.3 Research Questions

This paper asks four questions:

- **RQ1 (Behaviour):** How do DeepSeek and Gemini differ from humans in performance and observable choice behaviour across distinct IGT variants?

- **RQ2 (Mechanism):** Do LLM agents exhibit human-like learning and choice consistency dynamics as captured by Rescorla–Wagner (RW) parameters, including α, β, and negative log-likelihood (NLL), and do these dynamics change across variants?
- **RQ3 (Controllability):** Can prompt-engineered personas reliably induce distinct behavioural and cognitive phenotypes within each model, and does a sham persona fail to do so?
- **RQ4 (Mapping):** Where do LLM personas fall relative to human-derived RW clusters?

1.4 Contributions

This paper contributes a human-AI phenotype mapping framework that combines LLM-generated datasets across matched IGT variants with a RW-based mechanistic analysis under persona and sham controls.

2 Related Work

2.1 Decision-Making (IGT) and Cognitive Modelling (RW)

The IGT is a standard paradigm for studying decision-making under uncertainty [7]. Human behaviour in the IGT is heterogeneous, with individuals differing in how effectively they learn to favour advantageous options [8]. Large-scale studies, such as the Many Labs collaboration, demonstrate substantial variability, with some individuals consistently adopting optimal strategies while others persist with disadvantageous choices [9]. This dataset is also used in the present study as a human baseline for comparison.

To model such behaviour, reinforcement learning approaches are commonly applied, capturing how individuals update their choices based on feedback. Prior work has shown that reinforcement-learning-based models, including Expectancy Valence, win-stay/lose-shift, and Prospect Valence Learning (PVL), can explain human IGT behaviour [8]. In this study, the RW model is adopted as a simpler alternative, modelling learning as a trial-by-trial process driven by prediction error [10]. Behaviour can be summarised using interpretable parameters, including the learning rate (α) and choice consistency (β), enabling comparison of learning dynamics across agents. Although more complex models such as PVL are often used, RW is chosen for its simplicity and interpretability.

2.2 LLM Evaluation and the Gap in Behavioural Analysis

LLMs are typically evaluated using metrics based on accuracy and benchmark performance, limiting insight into how models make decisions over time [11]. Recent work argues that LLMs can be treated as participants in controlled experiments, allowing researchers to study how they learn, adapt, and respond to uncertainty over time rather than simply evaluating output quality [12]. However, such approaches remain relatively underexplored, leaving a gap in understanding the underlying decision-making dynamics of these models. This gap is further emphasised by the behavioural paradox identified by Suárez [13], where LLMs are capable of producing coherent and logically structured reasoning while simultaneously making irrational or inconsistent decisions in risky environments. These findings motivate behavioural evaluation frameworks rather than purely output-based ones.

2.3 Behavioural Analysis of AI Agents

Behavioural analysis has long been used in reinforcement learning to distinguish agents that achieve similar returns through different strategies [14]. This perspective extends to LLMs, which have been shown to exhibit diverse decision-making behaviours in structured tasks. Recent work demonstrates that LLMs can display varying tendencies such as over-exploration, over-exploitation, and inconsistent action selection depending on the task setup and prompting conditions [15]. These findings suggest that LLM behaviour is not fixed, but varies across contexts. Our study extends this line of work by introducing prompt-engineered personas as a controlled way to probe and influence behavioural variation, combining behavioural features, RW modelling, clustering, and a human baseline within a unified phenotype framework.

3 Task and Data

3.1 Iowa Gambling Task Environment

The IGT environment contains four decks (Q, K, S, D), each with distinct reward and penalty structures [16]. Each session begins with an initial total of 2000 units. On each trial, the agent selects a deck and receives a reward (win) along with a probabilistic penalty (loss), with the net outcome given by their sum. The payoff structure is as follows:

- Deck Q: reward of +100 with 50% probability of a −250 loss.
- Deck K: reward of +100 with 10% probability of a −1250 loss.
- Deck S: reward of +50 with 50% probability of a −50 loss.
- Deck D: reward of +50 with 10% probability of a −250 loss.

Decks Q and K are disadvantageous due to negative expected value, while S and D are advantageous, providing identical positive long-term returns but differing in loss frequency.

3.2 Human Dataset Description

Human behavioural data were obtained from the Psych-101 dataset, which is publicly available on the Hugging Face platform and contains IGT performance of humans across three experiments [17, 9]. Each session records trial-level choices and outcomes, with participants starting from an initial total of 2000. To ensure comparability, sessions are grouped into three variants based on task structure:

- Variant 1: 159 sessions with 100 trials.
- Variant 2: 323 sessions in total with mixed stopping rules (95, 100, or 150 trials).
- Variant 3: 29 sessions with 100 trials and an instruction to finish with at least 2500.

The data were then preprocessed to extract trial-level information, including experiment, session ID, trial number, deck choice, win amount, loss amount, and cumulative total.

3.3 Experimental Variants

The purpose of creating different variants is to study how various objectives and stopping constraints influence the decision-making process and learning strategies of humans and agents. Each variant uses the same underlying IGT environment but differs in the instructions given and the stopping rule.

Table 1. Experimental variants.

Variant	Trials	Goal	Stopping Rule
Variant 1	Exactly 100	Maximise profit	Stop at 100 trials
Variant 2	95, 100, or 150	Maximise profit	Stop when told
Variant 3	100	Finish with at least $2500	Stop when told

3.4 AI Agent Data Collection

AI agent data are collected using DeepSeek's `deepseek-chat` and Google's `gemini-3-flash-preview` APIs. For each variant, agents are given identical instructions to human participants and run across multiple independent sessions. The number of sessions and trials carried out is as follows:

- Variant 1: 200 sessions, 100 trials.
- Variant 2: 200 sessions, 100 trials.
- Variant 3: 50 sessions, 100 trials.

Persona-based experiments are conducted on Variant 2 to isolate prompting effects while holding the task framing constant. Six personas (risk-averse, risk-seeking, disengaged, exploratory, and two sham controls) were induced using prompt engineering for 50 sessions each. All collected data were then combined and stored in a standardised CSV format with columns such as session ID, trial number, deck choice, win amount, loss amount, final total, and persona label.

4 Methodology

This project investigates behavioural differences between human participants and LLM agents using the IGT, focusing on decision-making behaviour and underlying learning dynamics. As illustrated in Fig. 1, raw gameplay data are cleaned and standardised to ensure comparability. Analysis proceeds along two complementary pipelines. In the model-free pipeline, session-level behavioural features are extracted to capture observable decision-making patterns. In the model-based pipeline, a RW model is fitted to estimate learning parameters. These outputs are then clustered to identify behavioural phenotypes, with statistical tests used to evaluate persona effects and comparisons made with human data.

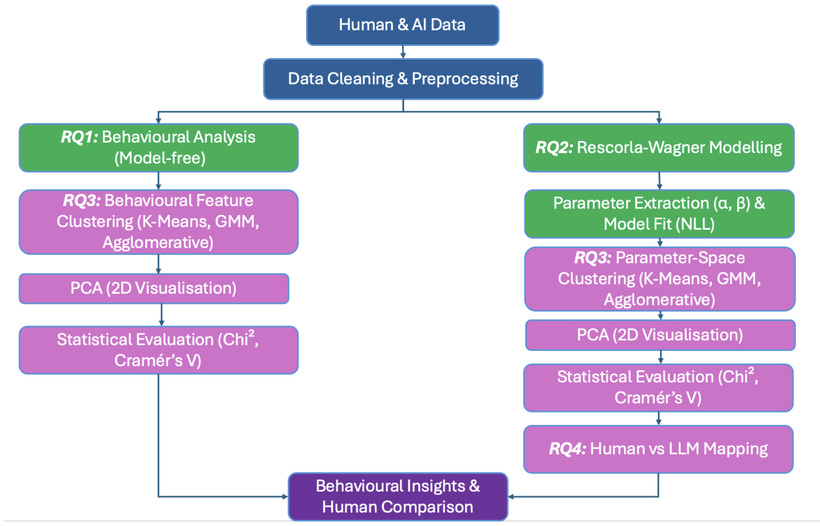


Fig. 1. Overview of analysis pipeline, illustrating the transformation of raw human and AI decision data into behavioural insights through modelling, parameter extraction, and clustering.

4.1 Experiment Design

The methodology addresses four linked questions: (1) model-free comparison of behavioural features across humans and LLMs; (2) RW fitting to estimate learning rate (α), choice consistency (β), and model fit (negative log-likelihood, NLL); (3) clustering using K-Means, Gaussian Mixture Models (GMM), and agglomerative clustering to test persona controllability; and (4) projection of human sessions into the LLM-derived RW space for phenotype mapping.

4.2 Behavioural Analysis

To address RQ1, a model-free behavioural analysis is conducted to compare human participants and LLM agents in the IGT, focusing on observable decision-making patterns without assuming underlying learning mechanisms [18]. To ensure comparability, analysis is restricted to fixed 100-trial sessions, with only the corresponding human experiment used to avoid confounding effects from variable stopping conditions. As human deck labels differ across datasets, choices are mapped within each session to generic actions (`action0`–`action3`), enabling consistent feature extraction across datasets. Session-level behavioural features include action proportions (`p_action0`–`p_action3`), mean reward, reward learning slope, switch rate, switch change, loss sensitivity, and final accumulated reward. These features capture performance, learning, exploration, and responses to losses. The features are analysed using descriptive statistics and visualisations, including reward trajectories, distribution plots, and action proportions. Action distributions are interpreted as indicators of behavioural concentration rather than specific deck preferences. This analysis provides a structured comparison of behavioural patterns across humans and LLM agents, forming the basis for subsequent modelling and clustering.

4.3 Rescorla–Wagner Modelling

To analyse how agents learn from experience in the IGT, we employ the RW model, a foundational framework for modelling trial-by-trial learning through prediction error [10]. At each trial t, the model updates the expected value V of the chosen action a using the prediction error, $R_t - V_t(a)$. The update rule is given by:

$$V_{t+1}(a) = V_t(a) + \alpha \left(R_t - V_t(a) \right), \tag{1}$$

where $\alpha \in (0, 1)$ represents the learning rate and R_t is the reward received at trial t.

Choice behaviour is modelled using a softmax function, where the inverse temperature parameter $\beta > 0$ controls the degree of choice consistency, with higher values indicating more deterministic behaviour. The parameters α and β are estimated for each session by minimising the NLL of the observed choices using L-BFGS-B optimisation. These parameters provide a low-dimensional representation of learning and decision-making behaviour, enabling direct comparison across human participants, LLM agents, and experimental conditions. This parameter space forms the basis for subsequent clustering and behavioural phenotyping (RQ3 and RQ4).

4.4 Clustering and Behavioural Phenotypes

Clustering is used to identify behavioural patterns across agents using both behavioural features and RW parameters (RQ3, RQ4) [20]. The optimal number of clusters is determined using silhouette score. In the behavioural feature space, clustering captures patterns in observable decision-making. Features are standardised for comparability and PCA is used for 2D visualisation. Clusters are not explicitly labelled due to high dimensionality, but are used to assess whether personas produce distinct groupings. In RW parameter space, clustering is applied to (α, β), providing an interpretable representation of learning behaviour with clusters labelled as adaptive, rigid, and extreme. Both analyses are performed on individual models and the combined dataset. K-means, GMM, and agglomerative clustering are compared using silhouette scores and adjusted Rand index, with K-means selected for stability and interpretability. Persona effects are evaluated using chi-square tests and Cramer's V. For RQ4, human RW parameters are projected into the LLM-derived K-means space, enabling direct comparison of behavioural distributions.

5 Results

5.1 Behavioural Results (RQ1)

Gemini consistently achieved the highest performance across all three variants (V1: $M = 3784.5$, SD $= 1024.9$; V2: $M = 3903.0$, SD $= 972.8$; V3: $M = 3869.0$, SD $= 1166.0$), followed by humans (V1: $M = 1901.9$, SD $= 926.1$; V2: $M = 1489.7$, SD $= 1311.8$; V3: $M = 1585.3$, SD $= 914.7$), with DeepSeek achieving the lowest performance (V1: $M = 1409.3$, SD $= 2129.3$; V2: $M = 1590.0$, SD $= 2305.1$; V3: $M = 1296.0$, SD $= 2552.7$) (Fig. 2A). However, outcome differences masked important behavioural differences. Humans, Gemini, and DeepSeek reached their results through distinct choice patterns rather than a shared strategy. Gemini highly favoured two actions (`action0` and `action3`) while almost completely ignoring others (`action2` ≈ 0.02). This suggests highly deterministic and exploitative behaviour (Fig. 3). Notably, this LLM illustrated clear and consistent reward learning while showing low loss sensitivity (≈ 0.27) across all variants (Fig. 2B).

Humans are more balanced explorers, with no strong bias toward specific card decks compared to both Gemini and DeepSeek. They displayed similar action distributions (approximately 0.25) across all three variants (Fig. 3). Humans displayed more variable and weaker learning patterns with difficulty adapting to various task constraints. Meanwhile, humans indicated a mean switch-after-loss rate of 0.68–0.77, showing a strong tendency to switch action after receiving a penalty. These findings suggest broader exploration and greater sensitivity to feedback. DeepSeek, with large negative outcomes, indicates instability in decision-making. Its action proportions showed inconsistent preferences, with `action0` being the most favoured choice. DeepSeek's rewards remained largely negative with minimal improvement over time, indicating weak learning. While it exhibited moderate loss sensitivity (0.47–0.61), this responsiveness did not translate into consistent or effective behavioural adjustment. LLM agents generally displayed a more skewed or unstable choice proportion compared to the balanced behaviour observed in humans. Humans adapt more strongly when dealing with loss compared to LLMs.

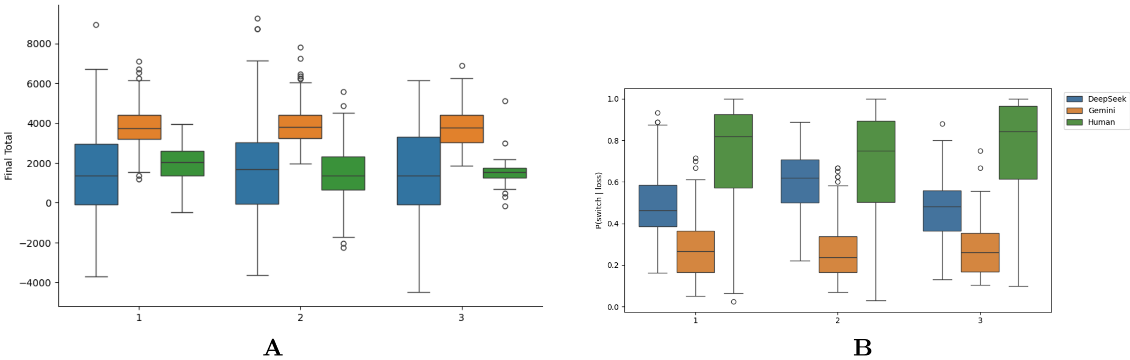


Fig. 2. A: Final Accumulated Reward of Humans and LLMs Across Variants **B:** Loss Sensitivity of Humans and LLMs Across Variants.

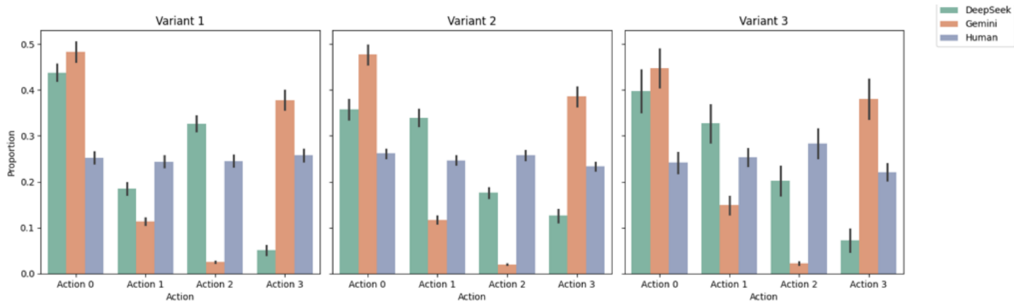


Fig. 3. Action selection behaviour of humans and LLMs across variants.

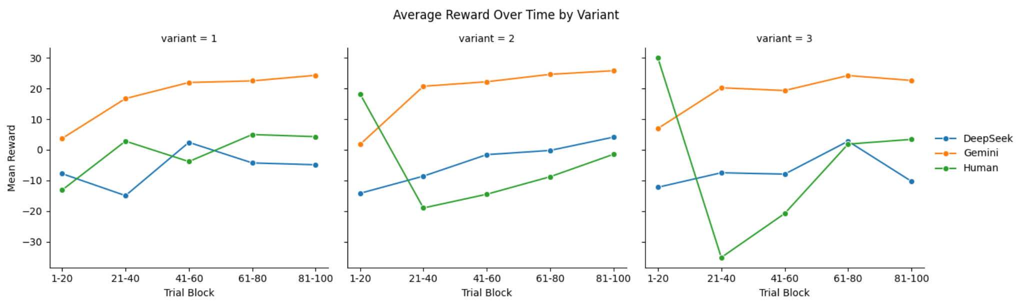
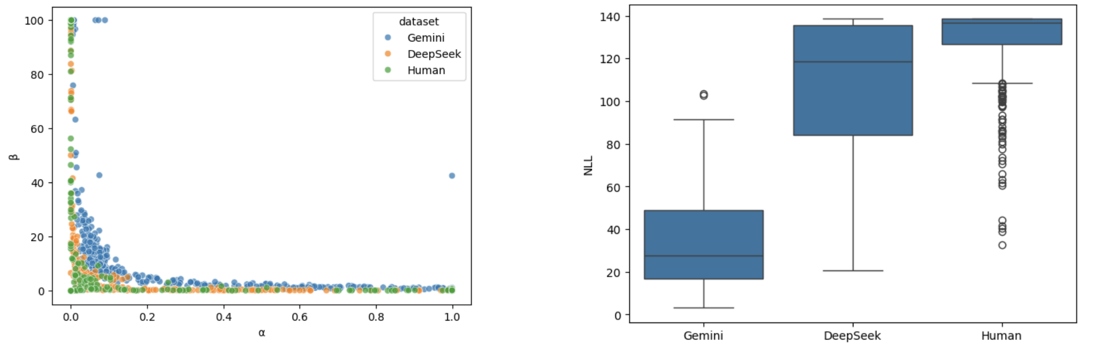


Fig. 4. Reward learning behaviour of humans and LLMs across variants.

5.2 Rescorla–Wagner Results (RQ2)

The relationship between α and β across datasets is shown in Fig. 5A. A clear inverse relationship is observed, where low α values are associated with high β, and higher α values correspond to lower β. This pattern appears across both humans and LLM agents, reflecting a similar trade-off captured by the RW model, rather than evidence of identical underlying learning mechanisms. Despite this shared structure, differences emerge in the distribution of parameters across datasets. Human behaviour is concentrated at very low α with a wide range of β values, including very high-β cases. Gemini shows a broader spread across parameter space with more moderate β values, while DeepSeek is more concentrated and closer to the human distribution but with fewer high-β cases.

Model fit results (Fig. 5B) further distinguish the datasets. Gemini achieves the lowest NLL values, indicating the best fit to the RW model. Human data show the highest NLL values, suggesting more complex behaviour that is less well captured by the model. DeepSeek lies between the two but exhibits higher NLL than Gemini, reflecting greater variability.



A **B**

Fig. 5. A: Relationship between learning rate (α) and choice consistency (β) across datasets. **B:** Comparison of Rescorla–Wagner model fit (NLL) across human and LLM datasets.

Overall, while LLMs reproduce the inverse α–β relationship seen in human behaviour, their parameter distributions and model fit indicate more structured and less variable decision-making.

5.3 Persona Effects (RQ3)

Behavioural Clustering. Two distinct behavioural feature clusters are identified using K-means (Fig. 6), representing different patterns of decision-making behaviour.

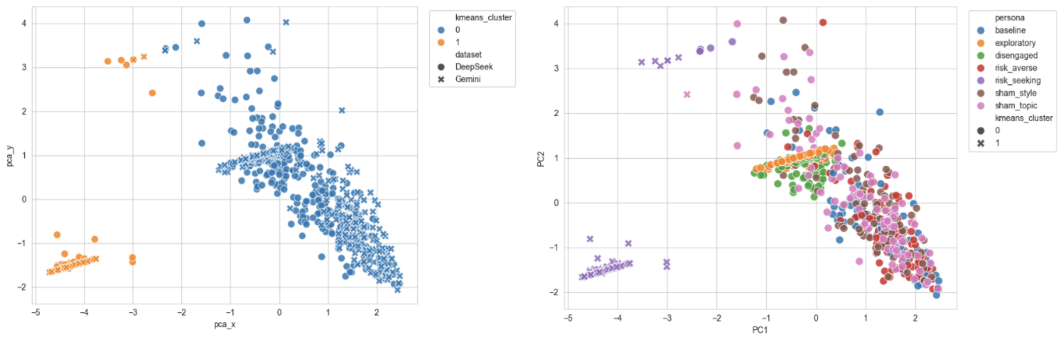


Fig. 6. PCA visualisation of behavioural feature clusters on the combined dataset and personas using K-means.

The clusters correspond to non-risk-seeking (Cluster 0) and risk-seeking (Cluster 1) behavioural phenotypes. As shown in Table 2, Cluster 0 is characterised by more distributed choices, positive learning trends, and higher final rewards, while Cluster 1 exhibits extreme concentration on deck Q, minimal learning, and consistently negative outcomes, reflecting a persistent risk-seeking strategy.

Table 2. Summary of behavioural feature means for K-means clusters in the combined dataset.

K-means Cluster	Deck Q	Deck K	Deck S	Net Score	Learning Slope	Risk Seeking	Loss Sensitivity	Final Total
0	0.1103	0.2280	0.3479	0.3236	0.1265	0.3382	0.5758	2733.94
1	0.9210	0.0726	0.0064	-0.9873	-0.0077	0.9936	0.0208	-495.83

Fig. 7 shows the distribution of persona conditions across clusters. Most personas are assigned to the non-risk-seeking cluster, including both sham controls, indicating that behavioural differences are limited and primarily driven by the risk-seeking persona. In contrast, the risk-seeking persona shows a strong shift towards Cluster 1, suggesting sensitivity to extreme behavioural biases.

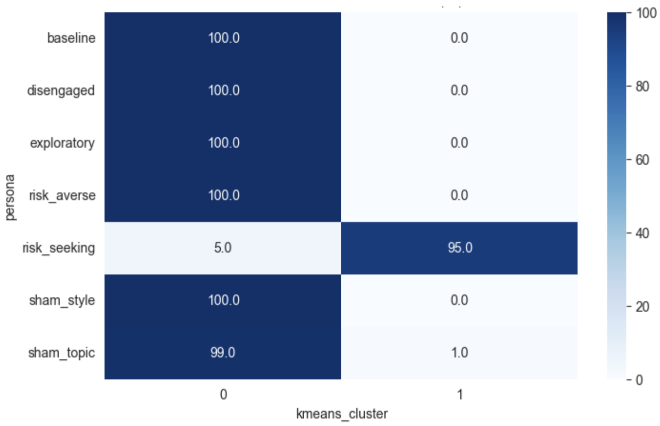


Fig. 7. Percentage distribution of persona conditions across non-risk-seeking and risk-seeking clusters.

The chi-square test results indicated a significant association between persona and cluster assignment ($\chi^2 = 651.49$, $p < 0.001$), with a very strong effect size (Cramer's $V = 0.965$). This indicates that persona prompting influences behaviour but primarily along a single dominant dimension separating risk-seeking from non-risk-seeking strategies.

Rescorla–Wagner Clustering. The clustering structure in RW parameter space is shown in Fig. 8. Data from Gemini and DeepSeek are combined and clustered using K-means ($k = 3$). K-means was selected based on silhouette analysis (0.756), outperforming agglomerative clustering (0.684) and GMM (0.150), indicating strong cluster separation.

Three clusters are identified: adaptive, rigid, and extreme behavioural phenotypes. Personas do not form distinct groupings, with all conditions overlapping within the same cluster structure.

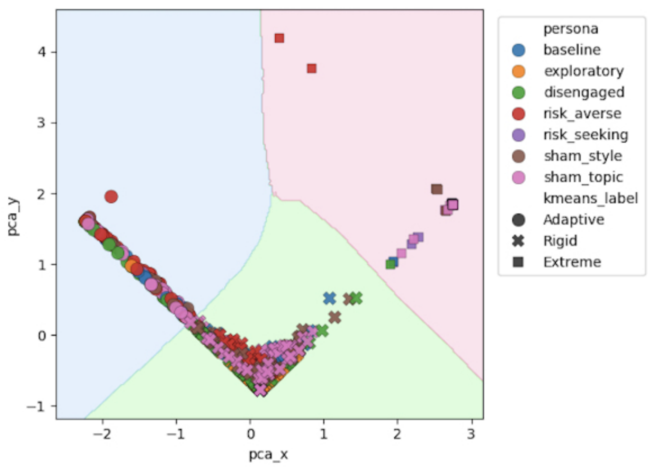


Fig. 8. PCA visualisation of RW parameter clusters using K-means.

Gemini shows a broader spread across clusters, particularly in the adaptive region, whereas DeepSeek is more concentrated within the rigid cluster. This indicates a shared behavioural structure but different distributions across models.

Table 3 summarises the distribution of personas across clusters. The rigid cluster dominates across all personas, reflecting low learning rates and consistent choice behaviour. However, differences emerge in cluster occupancy. Risk-seeking and exploratory personas are concentrated in the rigid cluster, while risk-averse shows a more balanced distribution between adaptive and rigid. Disengaged and sham personas exhibit mixed distribution with no distinct groupings.

Table 3. Percentage distribution of persona conditions across adaptive, extreme, and rigid clusters.

Persona	Adaptive	Extreme	Rigid
baseline	30	11	59
exploratory	3	0	97
disengaged	16	1	83
risk_averse	48	5	47
risk_seeking	0	11	89
sham_style	16	10	74
sham_topic	19	23	58

A chi-square test shows a significant association between persona and cluster assignment ($\chi^2 = 154.61$, $p < 0.001$), with a moderate effect size (Cramer's $V = 0.33$). This indicates that persona prompting has a measurable but constrained effect, shifting agents across existing clusters rather than creating new behavioural phenotypes.

5.4 Humans vs LLM Mapping (RQ4)

Human sessions were projected into the LLM-derived RW parameter space and assigned to existing clusters, enabling direct comparison of behavioural phenotypes. As shown in Table 4, human behaviour is primarily distributed across the rigid and extreme clusters, with low representation in the adaptive cluster. DeepSeek shows the closest alignment to this distribution, while Gemini differs with a higher proportion of adaptive behaviour and lower representation in the extreme cluster.

Table 4. Comparative distribution of human and LLM behaviour across Rescorla–Wagner parameter clusters (%).

Dataset	Adaptive	Extreme	Rigid
DeepSeek	11.14	10.29	78.57
Gemini	26.57	7.14	66.29
Human	12.61	21.62	65.77

A chi-square test indicates a significant association between dataset and cluster assignment ($\chi^2 = 57.79$, $p < 0.001$), though the effect size is small (Cramer's $V = 0.18$), suggesting modest differences. Standardised residuals show that Gemini is over-represented in the adaptive cluster, while humans are over-represented in the extreme cluster. DeepSeek shows smaller deviations, indicating closer alignment with human behaviour. Overall, DeepSeek more closely reproduces human-like behavioural phenotypes, while Gemini remains more distinct.

6 Discussion

6.1 Learning and Behavioural Strategy

Performance does not directly reflect underlying decision-making strategy. Gemini achieves the highest rewards through stable and consistent behaviour, indicating efficient but not human-like learning. Humans exhibit low learning rates with high choice consistency, reflecting rigid, experience-driven strategies that are not always optimal. DeepSeek shows a different failure mode, combining low learning rates with inconsistent choices, leading to unstable performance. All datasets share the same inverse relationship between learning rate (α) and choice consistency (β), indicating similar underlying learning dynamics. However, their distributions differ. Humans occupy a wider range, combining both exploratory and highly rigid behaviour. Gemini is more balanced and stable, while DeepSeek is closer to humans in learning rate but less consistent. Overall, LLMs capture the structure of human learning but not its variability. These results highlight a gap between performance and behaviour, where high performance can emerge from simple strategies rather than human-like decision-making.

6.2 Persona Controllability

Persona prompting demonstrates limited controllability. Behavioural clustering reveals a single dominant dimension separating risk-seeking from non-risk-seeking strategies, with only the risk-seeking persona forming a clearly distinct behavioural phenotype. All other personas cluster together, indicating that most prompting does not produce distinct behavioural patterns. In RW parameter space, personas shift the distribution of agents across existing clusters rather than creating new ones, indicating changes in underlying learning dynamics without consistent behavioural separation. Sham personas do not form distinct clusters and align with baseline behaviour, supporting their role as valid negative controls. This confirms that observed effects are driven by meaningful personality traits rather than incidental prompt variations.

6.3 Human Alignment

LLMs only partially align with human behaviour. While all datasets share the same parameter space, humans exhibit greater variability, spanning both highly exploratory and rigid strategies. DeepSeek is closest to this distribution but remains unstable, whereas Gemini is more constrained and less human-like. Overall, LLM behaviour is more structured and limited compared to humans.

7 Conclusion and Future Work

This study investigated the decision-making strategies and learning dynamics of LLMs compared to natural human cognition. Through a series of structured experiments under uncertainty in the IGT, we evaluated human participants with DeepSeek and Gemini agents across three task variants. To achieve this, we combined model-free behavioural analysis with a model-based RW reinforcement learning framework, which converts raw gameplay into latent cognitive parameters: learning rate and choice consistency. We also introduced prompt-engineered personas to each LLM to assess whether they could reliably induce distinct behavioural profiles and whether they align with human-like behavioural phenotypes.

The results showed that the LLM agents do not behave randomly in sequential decision-making tasks. Gemini achieved the highest task performance through stable and consistent behaviour, but did not exhibit human-like variability. Conversely, DeepSeek showed lower and more unstable performance, yet displayed a distribution of learning parameters closer to humans. While both LLMs replicate the inverse relationship between learning rate and choice consistency, they fail to capture the full variability of human decision-making. Persona prompting was found to have limited effect. Rather than creating new behavioural phenotypes, personas shifted agents within existing clusters, indicating that LLM behaviour is only weakly controllable through natural language instructions. These results highlight the effectiveness of behavioural phenotyping for assessing AI agents, especially in domains where it is just as important to know how a system makes a decision as it is to know what it does. However, these findings should be interpreted in light of the simplified RW model and the constrained IGT environment, which may not fully capture real-world decision-making complexity.

Future work should focus on employing more advanced cognitive models such as PVL to capture better variation of psychological traits. Analysis could be extended to more complex and non-stationary environments to evaluate whether the identified phenotypes adapt to hidden rule changes and non-stationary reward probabilities. Moreover, additional LLMs, such as ChatGPT and Claude, can be evaluated to improve robustness, and more sophisticated clustering methods could be acquired to reveal richer behavioural patterns and map AI agents better onto human phenotypes.

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